Week 4 · Lab 2

Version A

Graded Practice:

Operators and `if/else`

10 problems from LeetCode · 10 points each · checked by the TA at your desk

Week: 4 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lecture 3 — operators, short-circuit, `if/else if`, nested `if`, output formatting (`<iomanip>`)

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. There are **10 problems, 10 points each** (100 in total): problems 1–4 are **easy**, 5–7 **medium** and 8–10 **hard**. Solve them in any order; a problem with no attempt earns 0 points.
4. Under each title, **Practises:** says which part of Lecture 3 the problem trains.
5. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard (some are restricted to a fixed number of inputs, because loops and arrays come later). On LeetCode itself the answer is written as a *function* inside `class Solution`; you will learn functions in Week 6.

Quick reference (Lecture 3)

Arithmetic `17 / 5` is 3 and `17 % 5` is 2 (`int / int` throws the fraction away); `-7 / 2` is `-3`. `n % 10` is the last digit, `n / 10 % 10` the tens digit. Convert *before* dividing: `static_cast<double>(a) / b`. Compound forms: `x += 5; x %= 3;` `x++` / `++x` add 1.

Precedence, highest first: `()` · `!` · `* / %` · `+ -` · `< <= > >=` · `== !=` · `&&` · `||` · `=`. Inside `cout`, put a comparison in parentheses: `cout << (a < b);`

Conditions A range is two tests: `3 < x && x < 5` (`3 < x < 5` compiles but is always true). `&&` stops at the first false, `||` at the first true, so put the guard on the left:

```
if (c != 0 && a / c > 1) { ... }   // never divides by 0
```

Give long conditions a name, and remember that in arithmetic `true` counts as 1 and `false` as 0:

```
bool leap = (y % 4 == 0 && y % 100 != 0) || y % 400 == 0;
int odd = (a % 2 == 1) + (b % 2 == 1);   // 0, 1 or 2
```

Reading formatted input `cin >> c` for a `char` skips spaces and reads one character, so separators go into a spare `char`:

```
int y, m, d;  char dash;
cin >> y >> dash >> m >> dash >> d;     // 2019-02-10
```

`else if` chains The *first* true condition wins and the rest are skipped — test the strictest case first.

```
if (avg >= 90) {
    cout << "Excellent";
} else if (avg >= 75) {      // here avg < 90 already
    cout << "Good";
} else {
    cout << "Needs work";
}
```

Nested `if` An `else` binds to the *nearest* unmatched `if`. Always use braces.

Maximum, minimum, absolute value

```
int mx = a;
if (b > mx) { mx = b; }
if (c > mx) { mx = c; }      // mx = max(a, b, c)
int d = a - b;
if (d < 0) { d = -d; }       // d = |a - b|
```

Formatting output `#include <iomanip>`

```
cout << fixed << setprecision(2) << 2.5;   // 2.50
cout << setfill('0') << setw(3) << 7;      // 007
```

`fixed`, `setprecision` and `setfill` stay in force; `setw` applies to the *next item only*. Without `fixed`, `setprecision(2)` means 2 *significant* digits.

Classic bugs `if (x = 5)` (use `==`); `if (x > 0);` (stray `;`); `x == 6 || 28` (always true — write `x == 6 || x == 28`); `int` overflow (use `long long`).

EASY

Problems 1–4 · 40 points

Short programs: one idea each. Test every branch, not just the examples.

1 Account Balance After Rounded Purchase 10 pts

LeetCode 2806 · `leetcode.com/problems/account-balance-after-rounded-purchase`

Practises: `if/else` with `%`.

You have \$100. You buy something that costs `purchaseAmount` dollars ($0 \le$ `purchaseAmount` ≤ 100). The shop rounds the price to the nearest multiple of 10 (0 counts as one); if two multiples are equally close — the last digit is 5 — it rounds **up**. Print your balance after paying the rounded price.

Input	Output	Why
`9`	`90`	9 is rounded to 10.
`15`	`80`	10 and 20 are equally close; round up to 20.

TA: ____ / 10 pts initials _____

2 Power of Two 10 pts

LeetCode 231 · `leetcode.com/problems/power-of-two`

Practises: short-circuit `&&` as a guard.

Read an integer n (any `int`, negative values included). Print `true` if n is a power of two, that is $n = 2^x$ for some whole number $x \ge 0$; otherwise print `false`.

Input	Output	Why
`1`	`true`	$2^0 = 1$.
`16`	`true`	2^4.
`3`	`false`	

Hint: $2^{30} = 1073741824$ *is the largest power of two that fits in an* `int`*, and its only divisors are powers of two. Be careful with* $n = 0$ *and negative* n *— in which order must the two tests go?*

TA: ____ / 10 pts initials _____

3 Final Value of Variable After Performing Operations 10 pts

LeetCode 2011 · `leetcode.com/problems/final-value-of-variable-after-performing-operations`

Practises: `if/else` on a `char`.

A variable `X` starts at 0. Read **three** operations separated by spaces; each is one of `++X`, `X++` (add 1) or `--X`, `X--` (subtract 1). Print the final value of `X`.

Input	Output	Why
`--X X++ X++`	`1`	$0 - 1 + 1 + 1$.
`++X ++X X++`	`3`	

Hint: read each operation as three `char`*s. Which of the three characters always tells you the sign?*

TA: ____ / 10 pts initials _____

4 Airplane Seat Assignment Probability 10 pts

LeetCode 1227 · `leetcode.com/problems/airplane-seat-assignment-probability`

Practises: `if/else`, and `fixed` with `setprecision` from `<iomanip>`.

n passengers board a plane with exactly n seats, one at a time. The first passenger has lost the ticket and sits in a random seat. Every later passenger sits in their own seat if it is free, and otherwise in a random free seat. Read n ($1 \le n \le 10^5$) and print the probability that the **last** passenger sits in their own seat, with exactly 5 digits after the decimal point (as LeetCode does).

Input	Output	Why
1	`1.00000`	The only passenger gets the only seat.
2	`0.50000`	The first passenger takes seat 1 or seat 2, each with chance $\frac{1}{2}$.

Hint: work out $n = 3$ on paper: the first passenger takes seat 1, 2 or 3, each with chance $\frac{1}{3}$. What happens in each case?

TA: _____ / 10 pts initials _____

MEDIUM

Problems 5–7 · 30 points

A few cases to get right. Test the boundaries.

5 Maximum 69 Number 10 pts

LeetCode 1323 · `leetcode.com/problems/maximum-69-number`

Practises: an `else if` chain.

Read a positive integer `num` ($1 \le$ `num` ≤ 9999) whose digits are all 6 or 9. You may change **at most one** digit (a 6 into a 9 or a 9 into a 6). Print the largest number you can get.

Input	Output	Why
`9669`	`9969`	Changing the first 6 gives the biggest gain.
`9996`	`9999`	
`9999`	`9999`	Changing nothing is best.

Hint: which 6 should become a 9? Test the digits from the thousands down with an `else if` *chain; turning a 6 into a 9 in the hundreds place adds 300.*

TA: ____ / 10 pts initials _____

6 Minimum Amount of Time to Fill Cups 10 pts

LeetCode 2335 · `leetcode.com/problems/minimum-amount-of-time-to-fill-cups`

Practises: a maximum with `if`, then `if/else`.

A water dispenser fills cold, warm and hot water. Each second it fills either two cups of *different* types, or one cup of any type. Read how many cold, warm and hot cups are needed (each from 0 to 100) and print the minimum number of seconds.

Input	Output	Why
`1 4 2`	`4`	e.g. cold+warm, warm+hot, warm+hot, warm.
`5 4 4`	`7`	
`5 0 0`	`5`	Only one type: one cup per second.

Hint: two things slow you down: the largest amount (cups of one type are filled one per second), and the total (at most 2 cups per second). The answer is the larger of these two limits.

TA: ____ / 10 pts initials _____

7 Day of the Year 10 pts

LeetCode 1154 · `leetcode.com/problems/day-of-the-year`

Practises: an `else if` chain and a `bool` for the leap year.

Read a date written as `YYYY-MM-DD` (years 1900 to 2019, always a valid date) and print which day of its year it is. A year is a leap year (February has 29 days) if it is divisible by 4 but not by 100, or if it is divisible by 400.

Input	Output	Why
`2019-01-09`	`9`	
`2019-02-10`	`41`	31 + 10.

Hint: `cin >> y >> dash >> m >> dash >> d;` *with* `char dash`. *An* `else if` *chain on* `m` *gives the number of days in the months* before *it in a normal year (0, 31, 59, 90, . . .); add one more if the year is a leap year and* $m > 2$.

TA: ____ / 10 pts initials _____

HARD

Problems 8–10 · 30 points

More cases and more care. Read the hint before you start.

8 Distribute Money to Maximum Children 10 pts

LeetCode 2591 · `leetcode.com/problems/distribute-money-to-maximum-children`

Practises: nested `if`.

Read `money` ($1 \le$ `money` ≤ 200) and `children` ($2 \le$ `children` ≤ 30). All the money must be given out, every child gets at least \$1, and nobody gets exactly \$4. Print the largest possible number of children who get **exactly \$8**, or `-1` if the money cannot be given out under these rules.

Input	Output	Why
`20 3`	`1`	\$8, \$9, \$3. Two children with \$8 would leave \$4 for the third.
`16 2`	`2`	\$8 each.

Hint: first give everyone \$1. Each extra \$7 now turns one child into an \$8 child. Then handle the special cases: too much money left over, and the last child ending with exactly \$4.

TA: ____ / 10 pts initials _____

9 Day of the Week 10 pts

LeetCode 1185 · `leetcode.com/problems/day-of-the-week`

Practises: `else if` chains (day of the year, then the name).

Read a valid date as three numbers `day month year` (years 1971 to 2100) and print its day of the week: `Sunday`, `Monday`, `Tuesday`, `Wednesday`, `Thursday`, `Friday` or `Saturday`.

Input	Output	Why
`31 8 2019`	`Saturday`	
`18 7 1999`	`Sunday`	
`15 8 1993`	`Sunday`	

Hint: 1 January 1971 was a Friday. Count the days after it: 365 for every year from 1971 to `year - 1`*, plus one for every leap year among them, plus (day of the year − 1) — reuse problem 7. The number of leap years from 1 to p is* `p/4 - p/100 + p/400`*. Then* `(days + 5) % 7` *gives 0 = Sunday, 1 = Monday, . . . , 6 = Saturday.*

TA: ____ / 10 pts initials _____

10 Circle and Rectangle Overlapping 10 pts

LeetCode 1401 · `leetcode.com/problems/circle-and-rectangle-overlapping`

Practises: `else if` to push a value into a range.

Read a circle — `radius xCenter yCenter` — and an axis-parallel rectangle with bottom-left corner `x1 y1` and top-right corner `x2 y2`, all on one line in that order ($1 \le$ `radius` ≤ 2000, coordinates from -10^4 to 10^4). Print `true` if the circle and the rectangle have at least one point in common (touching counts), otherwise `false`.

Input	Output	Why
`1 0 0 1 -1 3 1`	`true`	They touch at $(1, 0)$.
`1 1 1 1 -3 2 -1`	`false`	
`1 0 0 -1 0 0 1`	`true`	

Hint: find the point of the rectangle closest to the centre. Its x is `xCenter` *pushed into the range* $[x_1, x_2]$*:* `x1` *if the centre is to the left,* `x2` *if to the right,* `xCenter` *otherwise — the same for y. Then compare squared distances; no square root is needed.*

TA: ____ / 10 pts initials _____

Score sheet — Version A

	Problem	Level	Points	Earned	TA
1	Account Balance After Rounded Purchase (LC 2806)	Easy	10		
2	Power of Two (LC 231)	Easy	10		
3	Final Value of Variable After Performing Operations (LC 2011)	Easy	10		
4	Airplane Seat Assignment Probability (LC 1227)	Easy	10		
5	Maximum 69 Number (LC 1323)	Medium	10		
6	Minimum Amount of Time to Fill Cups (LC 2335)	Medium	10		
7	Day of the Year (LC 1154)	Medium	10		
8	Distribute Money to Maximum Children (LC 2591)	Hard	10		
9	Day of the Week (LC 1185)	Hard	10		
10	Circle and Rectangle Overlapping (LC 1401)	Hard	10		
	Total		**100**		

After the lab

Every problem here is on LeetCode under the number shown. Some were cut down to a fixed number of inputs; with the loops of Week 5 you can solve the full versions. Once functions are covered (Week 6), rewrite your solutions as a LeetCode `class Solution` and submit them.

End of Week 4 · Lab 2 · Version A